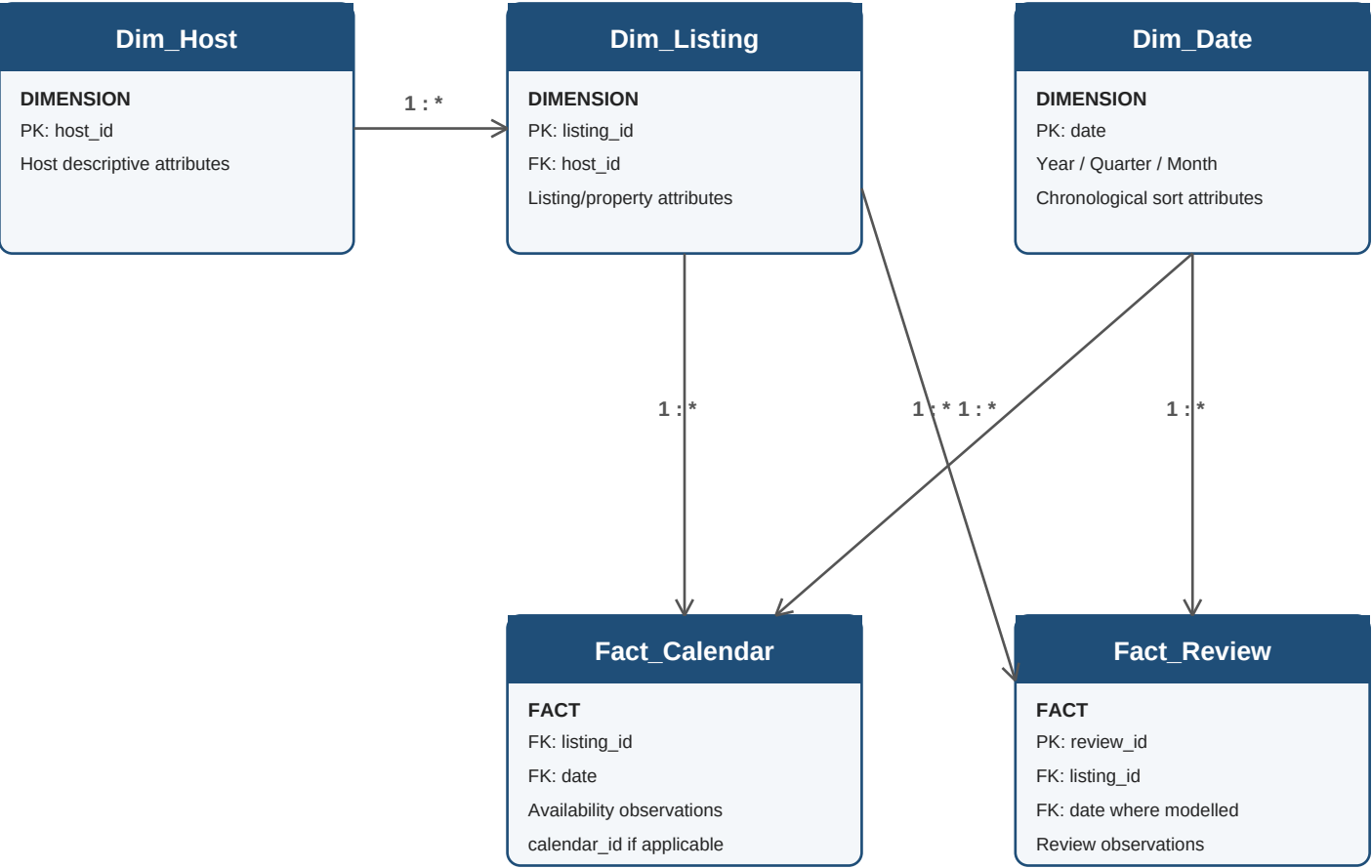


Airbnb Pricing Dashboard — Logical ERD

Target star-schema baseline. This page is on the star schema logical baseline. The source keys are the Model Data approval.

Airbnb Pricing Dashboard — Data ERD & Semantic Model Baseline

This document accompanies the visual ERD and defines the validation baseline for the semantic model.



Modelling intent

- Dimensions filter facts; prefer one-to-many, single-direction dimension → fact paths.
- Dim_Listing connects host attributes to listing-level calendar/review facts.
- Dim_Date connects reporting time to calendar observations and to review date where valid.
- Exact physical columns, data types, uniqueness and orphan status are validated during profiling.

2. Entity Register

Entity	Role	Purpose / Grain	Key
Dim_Host	Dimension	Host-level descriptive attributes	host_id
Dim_Listing	Dimension	Listing/property attributes; linked to host	listing_id
Dim_Date	Dimension	Reporting calendar/date attributes	date
Fact_Calendar	Fact	Calendar availability observations	listing_id + date (calendar_id if applicable)
Fact_Review	Fact	Review observations	review_id

3. Relationship Register

From	To	Cardinality	Relationship / intent
Dim_Host	Dim_Listing	1 : *	host_id — host filters listings
Dim_Listing	Fact_Calendar	1 : *	listing_id — listing filters calendar observations
Dim_Listing	Fact_Review	1 : *	listing_id — listing filters review observations
Dim_Date	Fact_Calendar	1 : *	date — date filters calendar observations
Dim_Date	Fact_Review	1 : *	date — only where a valid review date is modelled

4. Model Validation Requirements

Rule	Required validation
Grain	Confirm the grain of every source/model table before relationships are created.
Keys	Validate dimension-key uniqueness and fact foreign-key integrity.
Cardinality	Confirm each 1:* relationship against actual profiling results.
Filter direction	Use single-direction dimension → fact filtering wherever possible; avoid ambiguous paths.
Date table	Validate Dim_Date coverage and chronological sorting.
Technical fields	Hide technical keys/fields not intended for report consumers.
Reconciliation	Evidence Source → Raw → Clean → Model → DAX → Visual reconciliation before relevant gates pass.
Gate control	Do not move to the next phase until the current phase is reviewed and explicitly PASS/APPROVED.

5. Scope Controls

Control	Rule
Calendar unavailable	Do not interpret unavailable days as confirmed bookings/rented days.
Revenue	Confirmed/realised revenue and unsupported projected revenue are excluded.
Occupancy	Confirmed occupancy is excluded from the current model/report.
Scenario modelling	Occupancy/seasonal revenue What-If and dynamic price optimisation are excluded.
Causal inference	Observational findings must not be stated as causal conclusions.

Status: Visual ERD documented. The diagram is the logical target only; physical model implementation remains subject to source profiling and the Model Gate.